claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-abbcf063aff2c11ce.jsonl`
- SHA-256: `a6c6359aa2d5d619f5af542f8036d25a164e42fc9dc4c281f9d34d8f95a5adbf`
- Source modified: `2026-06-06T17:30:34+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:28:43.703Z)

Web Searcher: practitioner/local-levers

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Your angle: **practitioner/local-levers** ? Directly probes the two fully-local levers the user flagged as most promising (audio hit detection + pose/court-geometry serve-ready state machines) that run on-device with existing detectors.
Search query: `badminton shuttle hit detection audio onset thwack rally segmentation player pose serve stance court homography state machine`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.
Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.
Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-06T17:28:47.055Z)

I'll search for relevant resources on the two fully-local levers flagged as most promising.

3. user (2026-06-06T17:29:00.209Z)

Web search results for query: "badminton shuttle hit detection audio onset thwack rally segmentation player pose serve stance court homography state machine"

Links: [{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.org/pdf/2204.01899"},{"title":"[2204.01899] MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://ar5iv.labs.arxiv.org/html/2204.01899"},{"title":"A New Perspective for Shuttlecock Hitting Event Detection","url":"https://arxiv.org/html/2306.10293"},{"title":"[2306.10293] A New Perspective for Shuttlecock Hitting Event Detection","url":"https://ar5iv.labs.arxiv.org/html/2306.10293"},{"title":"A New Perspective for Shuttlecock Hitting Event Detection","url":"https://arxiv.org/pdf/2306.10293"},{"title":"TennisExpert: Towards Expert-Level Analytical Sports Video Understanding","url":"https://arxiv.org/pdf/2603.13397"},{"title":"Instant review system for badminton: computer vision system use case - Spyrosoft","url":"https://spyro-soft.com/blog/artificial-intelligence-machine-learning/instant-review-system-for-badminton-computer-vision-use-case"},{"title":"BFMD: A Full-Match Badminton Dense Dataset for Dense Shot Captioning","url":"https://arxiv.org/pdf/2603.25533"},{"title":"Automated Hit-frame Detection for Badminton Match Analysis","url":"https://arxiv.org/pdf/2307.16000"},{"title":"BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports","url":"https://arxiv.org/pdf/2502.21085"}]

Based on the search results, I found several relevant research papers on badminton analysis covering many of the topics in your query:

Key Research Areas Found:

Hit Detection & Rally Segmentation:
Identifying the shots of a rally is critical for reconstructing 3D trajectories, where a shot happens when a player hits the shuttle with their racket and ends right before the opposing player hits the shuttle or if the shuttle hits the court, making hit event identification equivalent to segmenting video into successive shots. Hit detection is formulated as a multi-class classification problem, with three-way classification for singles matches: no hit, near player hit, and far player hit predicted at each frame.

Player Pose & Court Information:
Hit detection models can be based on recurrent networks that consume court, pose, and 2D shuttlecock information to make hit predictions. MoveNet is a deep learning model designed for human pose estimation and tracking, specifically developed to accurately detect and track human body movements in real-time from video or camera input.

Audio-Based Approaches:

A lightweight, purely audio-based detection model can identify the distinct acoustic signatures of racquet-ball impacts from broadcast audio tracks.

The main papers identified are MonoTrack, "A New Perspective for Shuttlecock Hitting Event Detection," and "Automated Hit-frame Detection for Badminton Match Analysis" which address various aspects of your multi-component query.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

4. user (2026-06-06T17:29:10.960Z)

Web search results for query: "badminton rally segmentation audio shot detection deep learning ShuttleSet temporal action localization"

Links: [{"title":"VideoBadminton: A Video Dataset for Badminton Action Recognition | Request PDF ","url":"https://www.researchgate.net/publication/388099044_VideoBadminton_A_Video_Dataset_for_Badminton_Action_Recognition"},{"title":"Bridging the Gap: Doubles Badminton Analysis with Singles-Trained Models","url":"https://arxiv.org/html/2508.13507v1"},{"title":"FineBadminton: A Multi-Level Dataset for Fine-Grained Badminton Video Understanding","url":"https://arxiv.org/html/2508.07554v1"},{"title":"Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges","url":"https://arxiv.org/html/2505.03991v3"},{"title":"[PDF] ShuttleNet: Position-aware Fusion of Rally Progress and Player Styles for Stroke Forecasting in Badminton | Semantic Scholar","url":"https://www.semanticscholar.org/paper/ShuttleNet:-Position-aware-Fusion-of-Rally-Progress-Wang-Shuai/9145c3426b54be4ab7013d20fc584be76d2e2d58"},{"title":"FineBadminton: A Multi-Level Dataset for Fine-Grained ...","url":"https://arxiv.org/pdf/2508.07554"},{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}]

Based on the search results, I found several relevant resources on badminton analysis using deep learning methods:

Key Findings:

ShuttleSet Dataset: ShuttleSet is a broadcasting badminton dataset with documented shot frames that mark the moment when the racket hits the shuttlecock, and it has been used as a benchmark for badminton action recognition research.

Deep Learning Approaches for Shot Detection: Recent methods utilize VideoMAE fine-tuned on badminton videos to extract visual features for identifying hit events, which are then combined with ball and player coordinate data through spatio-temporal feature learning modules to classify stroke types and localize shuttlecock landing spots.

Rally Segmentation and Temporal Action Localization: Temporal action localization (TAL) is effective for analyzing long or continuous actions such as rallies in racket sports. Previous badminton datasets like ShuttleNet, ShuttleSet, and ShuttleSet22 have primarily focused on stroke detection and rally localization.

Shot Detection Methods: TrackNet is a deep neural network designed for tracking small objects like shuttlecocks, and when combined with YOLOv7 for player swing identification, a shot refinement algorithm can be used to obtain the correct hitting moment.

The research shows active development in combining visual features, temporal modeling, and deep learning architectures for comprehensive badminton video understanding.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. assistant (2026-06-06T17:29:14.448Z)

Let me dig into the two most directly relevant papers for the local-levers angle (audio hit detection and pose/court state machines) plus the hit-event detection paper.

6. user (2026-06-06T17:29:25.632Z)

Web search results for query: "tennis table tennis ball hit detection audio onset spectrogram CNN sound impact rally point segmentation"

Links: [{"title":"(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis","url":"https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"},{"title":"(PDF) Detection of ball hits in a tennis game using audio and visual information","url":"https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"},{"title":"Spin Detection Using Racket Bounce Sounds in Table Tennis This research was funded by Sony AI.","url":"https://arxiv.org/html/2409.11760v1"},{"title":"Spin Detection Using Racket Bounce Sounds in Table Tennis","url":"https://arxiv.org/pdf/2409.11760v1"},{"title":"Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline This work was funded by Sony AI","url":"https://arxiv.org/html/2409.11760"},{"title":"Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/"}]

Here are the search results for your query about tennis/table tennis ball hit detection using audio analysis with spectrograms and CNNs:

Key Findings:

Audio-Based Ball Hit Detection Approaches:

Researchers detect high-frequency peaks that may correspond to table tennis ball bounces, then refine these results using a CNN-based classifier that accurately predicts both the type of racket used and whether spin was applied.

Spectrogram and CNN Methods:

The preprocessing involves audio sampled at 44.1 kHz with standardized length of 661 samples (15 ms), converting stereo to mono and transforming into a Mel spectrogram, with the CNN performing best using 64 Mel bands and an FFT window length of 256 samples, with Mel spectrograms consistently producing the best results.

Sound Event Detection and Temporal Accuracy:

Bounce candidates are detected using energy peaks based on an exponential moving average on high-pass filtered signals, converted to Mel spectrograms and classified using CNNs to identify bounce surfaces and detect spin, achieving millisecond-level onset accuracy.

Tennis Impact vs. Rebound Analysis:

Impact events exhibit sharp, powerful energy distribution with higher magnitude in lower frequencies than rebounds, while rebounds show more prolonged and less intense energy distribution.

The research demonstrates that modern pipelines tackle bounce detection, surface classification (racket, table, floor), and spin detection from audio, supported by datasets of thousands of annotated bounce recordings.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-06-06T17:29:27.202Z)

Shuttlecock Hitting Event Detection Summary

Method Overview

The approach combines multiple deep learning models to detect badminton hitting events from video sequences rather than relying on generic methods.

Key Components

Models Used:
- **SwingNet** (MobileNetV2 + bidirectional LSTM): Detects hitting frames and shot sequences from optical flow videos
- **Vision Transformer (ViT-B/16)**: Classifies player attributes (hitter identity, swing type, ball height, ball type, match winner)
- **YOLOv5m**: Detects player and ball positions
- **YOLOv8x-pose-p6**: Estimates player poses for accurate position detection
- **TrackNetV2**: Supplements ball trajectory tracking

Information Sources:
The method uses pose estimation and ball trajectory but notably does NOT utilize court information or audio.

Data & Labels Required

Training requires annotated video clips with frame-level labels for: hitting frames, player identities, swing characteristics, ball positions, and landing coordinates.

Reported Performance

Maximum score achieved: **0.0727 out of 1.111** (approximately 6.5%), with the primary limitation being SwingNet's keyframe detection reaching only "0.0426" of its potential 0.2 maximum.

Code & Licensing

The authors state that "source code will be available" at a GitHub repository, but **no dataset or code license is explicitly mentioned** in this paper.

Local Runnability

Not easily locally runnable?the method requires 64 consecutive 180×180 frames simultaneously in memory, demanding substantial GPU resources (author notes their 6GB GPU was limiting).

8. assistant (2026-06-06T17:29:32.479Z)

Let me confirm details on the two most directly actionable, locally-runnable papers: the audio-based tennis hit detection and the MonoTrack pose/court hit-detection approach.

9. user (2026-06-06T17:29:41.245Z)

MonoTrack: Hit Detection and Rally Segmentation

Detection Approach

MonoTrack leverages multiple signals to detect hits. The system uses "court layout, the poses, and the shuttlecock location" to identify when players strike the shuttle. These three components?court homography, player poses from pose estimation, and 2D shuttle tracking?are combined in a unified framework.

Hit Detection Architecture (HitNet)

The hit detection uses a recurrent neural network called HitNet with this structure:

- **Input features**: Normalized pixel coordinates of court corners, both players' poses, and shuttle location
- **Processing**: Features embedded into 32 dimensions, then fed through two GRU layers processing 12-frame windows

- **Output**: Three-class classification (no hit, near player hit, far player hit)
- **Size**: Approximately 16K parameters, enabling inference at "several thousand frames a second"

Accuracy Results

The system achieves strong performance on hit detection:

- **Standalone model**: 86.3% accuracy
- **With constrained optimization**: "94.3% recall, 94.9% precision, and 0.946 F1 score"

The optimization applies domain knowledge constraints (hit frequency, minimum spacing, player alternation) to refine raw predictions.

Code and Licensing

The paper does not mention code availability, licensing terms, or GPU requirements. No information about reproducibility on consumer hardware is provided in this document.

10. user (2026-06-06T17:29:43.877Z)

Web search results for query: "tennis match point rally segmentation audio crowd applause silence broadcast highlight extraction local"

Links: [{"title":"TennisExpert: Towards Expert-Level Analytical Sports Video Understanding","url":"https://arxiv.org/pdf/2603.13397"},{"title":"An intelligent strategy for the automatic detection of highlights in tennis video recordings - ScienceDirect","url":"https://www.sciencedirect.com/science/article/abs/pii/S0957417409001146"},{"title":"TennisExpert: Towards Expert-Level Analytical Sports Video Understanding","url":"https://arxiv.org/html/2603.13397v2"},{"title":"[PDF] Detection of Tennis Events from Acoustic Data | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Detection-of-Tennis-Events-from-Acoustic-Data-Baughman-Morales/5757cc038c864dc0f2d676794faaa4ff8961a97f"},{"title":"Crowd noise analysis","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/8457768"},{"title":"HMM BASED STRUCTURING OF TENNIS VIDEOS USING VISUAL AND AUDIO CUES","url":"https://www.irisa.fr/metiss/gravier/biblio/03/kijak-icme03.pdf"},{"title":"AUDIOVISUAL INTEGRATION FOR TENNIS BROADCAST STRUCTURING","url":"https://www.irisa.fr/metiss/publis/kijak-cbmi03.pdf"},{"title":"Leveraging Contextual Cues for Generating Basketball Highlights","url":"https://arxiv.org/pdf/1606.08955"},{"title":"EVENT DETECTION IN TENNIS MATCHES BASED ON VIDEO DATA MINING","url":"https://mmcv.csie.ncku.edu.tw/~wtchu/papers/2008ICME-tien.pdf"},{"title":"Playing sports videos using the audio-based highlights detection. At... | Download Scientific Diagram","url":"https://www.researchgate.net/figure/Playing-sports-videos-using-the-audio-based-highlights-detection-At-the-top-the-audio_fig2_228361562"}]

Here are the search results for your query about tennis match point rally segmentation, audio, crowd applause, silence, broadcast highlight extraction, and local analysis:

Key Findings:

Rally Segmentation Using Audio:
Professional tennis broadcasts typically consist of alternating periods of rally play and pauses between points, and to organize datasets at the rally level, full match broadcasts are segmented into individual rally clips. To achieve this efficiently across hundreds of hours of video, a lightweight, purely audio-based detection model is employed that trains a neural network to identify the distinct acoustic signatures of racquet-ball impacts from the broadcast audio track.

Audio Feature Robustness:
To ensure the model is robust against complex ambient stadium noises, training data is augmented by overlaying various background sounds (e.g., crowd cheering, clapping, and announcer voices) onto the audio samples, allowing the model to reliably distinguish true hitting events from similar impulsive noises in the broadcast.

Highlight Detection Using Audio Cues:
Valuable information is carried in multiple channels of the media such as text, audio (applause/cheering), and speech that accompany the visual components. Among these, the main cue lies in detecting the intensity of the commentator and the audience applause.

Crowd Noise Analysis:
An audio stream for an event containing crowd noise is captured and time coded, with regions identifying crowd noise based on spikes in audio level, serving as a gauge of crowd reaction to events occurring during the event.

Audio-Visual Integration:
One single stream cannot provide enough information to identify different scenes. Research indicates that combining audio and visual features improves segmentation accuracy for rally detection and event classification in tennis broadcasts.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-06T17:29:48.415Z)

I have strong coverage of both local-levers. The MonoTrack HitNet result is especially relevant (pose + court homography + shuttle trajectory state machine, 0.946 F1, tiny 16K-param model). Let me do one more targeted check on ShuttleSet licensing since the user explicitly asked about datasets.

12. user (2026-06-06T17:29:55.142Z)

Web search results for query: "ShuttleSet badminton stroke dataset license GitHub CC BY commercial use TrackNetV2 dataset license"

Links: [{"title":"SENTATION FUSION NETWORK FOR PRE-STROKE PRE","url":"https://openreview.net/pdf/e360f0b2cb53ae8733ca7f6703925a362e9d94bd.pdf"},{"title":"(PDF) ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis","url":"https://www.researchgate.net/publication/371413847_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"},{"title":"[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis","url":"https://arxiv.org/abs/2306.04948"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining","url":"https://dl.acm.org/doi/10.1145/3580305.3599906"},{"title":"MonoTrack: Shuttle trajectory reconstruction from monocular badminton video","url":"https://arxiv.org/pdf/2204.01899"},{"title":"BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports","url":"https://arxiv.org/html/2502.21085v2"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Request PDF","url":"https://www.researchgate.net/publication/372910678_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"},{"title":"ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton   Tactical Analysis","url":"https://arxiv.org/pdf/2306.04948"},{"title":"Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset","url":"https://arxiv.org/pdf/2306.15664"}]

Based on my search results, I found information about these badminton datasets:

ShuttleSet Dataset:
The ShuttleSet dataset and corresponding benchmarks are available at https://github.com/wywyWang/CoachAI-Projects. It contains 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021 with 27 top-ranking men's singles and women's singles players.

TrackNetV2 Dataset:
TrackNetV2 is a public dataset that contains 77k annotated frames from 26 unique singles matches from international play, filmed from an overhead, static, "broadcast-view" camera.

However, the search results I found did not contain specific information about the CC BY commercial use licenses for either of these datasets. While the results confirm that both datasets are publicly available and hosted on GitHub, they do not provide details about their specific licensing terms (such as CC BY or restrictions on commercial use). To get accurate license information, you would need to check the GitHub repositories directly or consult the official dataset documentation.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. assistant (2026-06-06T17:30:19.868Z)

I have enough to deliver a focused, ranked set. The MonoTrack/HitNet paper is the strongest hit for the local-levers angle (pose + court homography + shuttle state machine, 0.946 F1, ~16K params ? runs trivially on the RTX 2070 Super), followed by the audio hit-detection papers (table tennis Mel-spectrogram CNN with code, and the tennis acoustic onset work), plus the broader shuttlecock-hitting-event survey and ShuttleSet dataset for labels.

14. user (2026-06-06T17:30:31.413Z)

Structured output provided successfully

15. assistant (2026-06-06T17:30:34.595Z)

Returned 6 ranked results focused on the two local-first levers (audio hit detection + pose/court-geometry state machines), with MonoTrack/HitNet as the strongest match.